

CHARLEVOIX, MI, USA

WMO: 727434

Lat: 45.305N

Lon: 85.275W

Elev: 197

StdP: 98.98

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 04870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.6	-15.1	-21.9	0.5	-16.9	-19.0	0.7	-13.9	12.9	-6.2	11.7	-6.2	2.8	100	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.1	21.3	27.7	20.6	26.4	19.9	22.9	27.1	22.0	25.9	21.2	24.8	4.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.7	25.1	20.9	16.0	24.3	19.9	15.0	23.7	68.7	27.2	65.4	25.9	62.4	24.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.5	8.5	DB	-22.4	32.7	4.0	1.7	-25.2	33.9	-27.6	34.9	-29.8	35.9	-32.7	37.1	(4)
(5)			WB	-22.7	24.5	3.9	0.9	-25.5	25.1	-27.8	25.7	-29.9	26.2	-32.7	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-5.4	-5.6	-1.1	5.1	11.5	16.9	20.2	19.9	16.6	9.6	3.4	-1.8	(6)
(7)		DBStd	10.28	5.13	5.32	6.18	4.85	4.67	3.97	3.12	2.84	4.05	4.48	4.30	4.29	(7)
(8)		HDD10.0	2093	476	437	348	162	36	1	0	0	2	64	203	365	(8)
(9)		HDD18.3	4155	734	671	602	397	221	72	16	16	80	274	449	623	(9)
(10)		CDD10.0	1191	0	0	4	15	82	209	318	308	199	51	4	0	(10)
(11)		CDD18.3	209	0	0	0	0	8	30	76	65	28	2	0	0	(11)
(12)		CDH23.3	1338	0	0	3	6	67	233	507	344	161	16	0	0	(12)
(13)		CDH26.7	264	0	0	0	1	14	56	106	57	31	1	0	0	(13)
(14)	Wind	WSAvg	3.8	4.7	4.4	3.7	3.9	3.3	2.9	2.9	3.1	3.4	4.2	4.6	4.8	(14)
(15)	Precipitation	PrecAvg	824	61	40	45	60	75	74	75	83	86	91	72	61	(15)
(16)		PrecMax	956	110	69	112	109	131	163	127	148	163	196	123	123	(16)
(17)		PrecMin	694	25	13	7	13	23	13	38	23	22	37	30	11	(17)
(18)		PrecStd	79	24	16	26	24	28	36	28	34	33	42	27	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	9.1	20.8	23.8	28.6	31.2	31.7	30.5	29.8	25.3	18.4	12.5	(19)
(20)			MCWB	6.1	6.5	14.6	14.2	19.5	22.8	23.1	21.8	21.6	19.0	13.0	11.0	(20)
(21)		2%	DB	5.2	5.9	13.7	19.1	25.0	27.9	29.0	28.0	27.3	22.3	14.1	7.1	(21)
(22)			MCWB	3.4	3.3	9.9	12.7	17.8	20.4	21.5	21.2	20.1	17.4	10.8	5.3	(22)
(23)		5%	DB	3.0	2.9	10.2	16.2	22.2	26.1	27.7	26.8	25.1	19.2	12.0	5.4	(23)
(24)			MCWB	1.8	1.3	6.8	9.9	16.0	19.2	20.9	20.5	19.5	15.5	9.1	4.1	(24)
(25)		10%	DB	1.8	1.3	7.1	12.8	19.0	23.8	26.2	25.2	22.9	16.5	9.9	3.8	(25)
(26)			MCWB	1.0	0.0	4.0	7.8	14.4	18.3	20.1	19.6	18.6	13.4	7.7	2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	7.3	14.9	16.3	20.9	23.5	24.5	23.5	23.1	20.8	14.1	10.5	(27)
(28)			MCDWB	7.4	8.8	20.1	20.7	26.5	29.1	29.6	27.7	26.7	23.5	17.3	12.0	(28)
(29)		2%	WB	3.7	3.5	10.7	13.3	19.1	21.7	23.1	22.5	21.7	18.4	11.3	5.9	(29)
(30)			MCDWB	5.0	5.3	12.8	18.4	23.5	26.8	27.1	26.4	25.2	21.1	13.8	6.6	(30)
(31)		5%	WB	2.0	1.6	6.7	10.6	17.1	20.3	22.1	21.7	20.6	15.7	9.5	4.0	(31)
(32)			MCDWB	3.0	3.0	9.7	15.3	20.9	24.5	25.8	25.2	23.9	18.9	11.8	5.1	(32)
(33)		10%	WB	0.8	0.1	4.3	8.3	15.0	19.1	21.2	20.7	19.4	13.7	7.6	2.7	(33)
(34)			MCDWB	1.9	1.0	6.8	12.6	18.5	22.9	24.9	23.8	22.5	16.3	9.4	3.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	6.7	8.3	9.0	10.3	9.7	9.2	8.5	8.9	7.2	5.8	4.6	(35)
(36)			MCDBR	6.7	8.1	12.1	14.9	14.0	12.2	11.0	10.3	11.1	11.6	9.6	7.0	(36)
(37)			MCWBR	6.0	6.7	8.5	9.0	7.8	6.3	5.6	5.3	5.9	7.3	7.0	6.4	(37)
(38)		5% WB	MCDWB	6.3	7.9	11.5	14.1	12.7	10.4	9.1	8.4	9.5	10.3	8.9	6.6	(38)
(39)			MCWBR	5.7	6.9	8.5	9.4	8.2	6.4	5.4	5.1	6.1	7.5	7.1	6.4	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.291	0.293	0.318	0.351	0.382	0.392	0.389	0.383	0.371	0.340	0.304	0.280	(41)	
(42)		taud	2.375	2.345	2.382	2.357	2.348	2.351	2.387	2.404	2.420	2.484	2.518	2.464	(42)	
(43)		Ebn at Noon	819	891	915	909	887	876	874	867	847	829	807	802	(43)	
		Edh at Noon	74	92	103	116	121	121	116	110	99	81	65	62		
(44)	All-Sky Solar Radiation	RadAvg	1.17	2.10	3.55	4.61	5.63	6.12	6.16	5.24	4.02	2.35	1.26	0.86	(44)	
(45)		RadStd	0.13	0.21	0.37	0.47	0.48	0.44	0.24	0.33	0.34	0.28	0.18	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page